

Online Exam Portal

Test Plan

Software Engineering Mini-Project — Deliverables Part-1

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Revision History

Version	Date	Author	Description
1.0	28-Sep-2026	[Student Name(s)]	Initial release of the Test Plan with test cases and traceability to SRS OEP-SRS-001.

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1. Introduction

1.1 Test Plan Identifier

OEP-TP-001, version 1.0. This plan is derived from the Software Requirements Specification OEP-SRS-001 and the Software Architecture & Design Specification OEP-SADS-001.

1.2 Purpose

This plan defines the scope, approach, resources and schedule of the testing activities for the Online Exam Portal (OEP). Its objectives are to verify that every requirement in the SRS is implemented correctly, to expose defects early and to provide evidence that the system is secure, reliable and performs adequately during an examination.

1.3 Scope

Testing covers all functional requirements (FR-01 to FR-21), non-functional requirements (NFR-01 to NFR-10) and security requirements (SR-01 to SR-10) of the SRS for release 1.0 of the OEP web application and REST API.

1.4 References

- OEP-SRS-001 — Software Requirements Specification v1.0.
- OEP-SADS-001 — Software Architecture & Design Specification v1.0.
- IEEE Std 829-2008 — Standard for Software and System Test Documentation.
- OWASP Application Security Verification Standard 4.0; OWASP Testing Guide v4.2.

1.5 Definitions

Term	Meaning
TC	Test Case
UAT	User Acceptance Test
DAST / SAST	Dynamic / Static Application Security Testing
Defect severity	Critical: system unusable or data loss; High: major feature broken; Medium: feature partly broken with workaround; Low: cosmetic

2. Test Plan Overview

2.1 Test Levels

Level	Objective	Responsible	Basis
Unit testing	Verify individual classes/functions (target coverage $\geq 70\%$, NFR-08).	Developers	Design specification
Integration testing	Verify interaction between services, API and database.	Developers / QA	Component diagram, API design
System testing	Verify complete system against the SRS (test cases in Section 12).	QA team	SRS
Acceptance testing	Confirm the system meets user needs in a mock exam session.	Instructor / representatives	SRS, use cases

2.2 Entry and Exit Criteria

Entry: the SRS and design are approved; the build is deployed to the test environment; smoke tests pass; test data is loaded.

Exit: 100 % of High-priority test cases executed; at least 95 % of all test cases passed; no open Critical or High defects; all security test cases (Section 5.1) passed.

3. Test Items

The following items are subject to testing. Version identifiers follow the format MAJOR.MINOR.BUILD.

Item	Description	Version under test
Web client	React single-page application (student, teacher and admin screens).	1.0.x
Authentication Service	Registration, login, logout, token issue, password reset.	1.0.x
User Management Service	Administrator user and role management.	1.0.x
Question Bank Service	CRUD of questions.	1.0.x
Exam Management Service	Exam creation, publishing and assignment.	1.0.x
Exam Attempt Service	Start, timer, autosave and submission of attempts.	1.0.x
Evaluation Service	Automatic marking and score computation.	1.0.x
Result & Report Service	Results, statistics and CSV export.	1.0.x
Audit & Logging Service	Recording of security events.	1.0.x
Database schema	PostgreSQL schema and constraints.	1.0.x

4. Features to be Tested

4.1 Features in Scope

Feature area	Requirements	Test cases
Registration, login, logout, password reset	FR-01 to FR-04	TC-01 to TC-04
User management	FR-05	TC-17
Exam and question-bank management	FR-06 to FR-09	TC-05, TC-06, TC-18
Exam taking, timer, autosave, navigation	FR-10 to FR-15, FR-20	TC-07, TC-08, TC-15, TC-18, TC-21
Evaluation and results	FR-16, FR-17	TC-09
Reports	FR-18, FR-19	TC-20
Audit logging	FR-21	TC-19
Performance and scalability	NFR-01, NFR-02, NFR-04	TC-10
Usability, compatibility, accessibility	NFR-05, NFR-06, NFR-10	TC-14
Reliability	NFR-07	TC-15
Availability, maintainability, recoverability	NFR-03, NFR-08, NFR-09	Inspection (Section 5.5)
Security	SR-01 to SR-10	TC-03, TC-08, TC-11, TC-12, TC-13, TC-16, TC-19, TC-22, TC-23

4.2 Features not to be Tested

- Subjective/essay grading, live proctoring and payment features (out of scope of release 1.0).

- Third-party components (PostgreSQL, Redis, SMTP server) beyond verifying correct integration.
- Native mobile apps and LMS integration (not part of the product).

5. Test Approach

Testing follows a risk-based approach: features that affect exam fairness, data integrity and security receive the deepest coverage. Test cases are written once (Section 12), executed manually first and then automated where stable (API tests with Postman/Newman, UI tests with Selenium).

5.1 Security Validation

Security validation demonstrates that the security requirements SR-01 to SR-10 of the SRS and the security architecture of the design specification are correctly implemented. It combines static review, dynamic testing and configuration inspection.

Security area	Validation activity	Tool / method	Requirement	Test case	Pass criterion
Authentication	Brute-force, lockout and rate-limit checks	Postman, JMeter	SR-06, FR-03	TC-03	Lock after 5 failures; HTTP 429 on excess requests
Password storage	Inspect stored hashes and code review	DB query, code review	SR-01	TC-16	bcrypt hash, cost ≥ 12 , no plaintext
Transport security	TLS configuration scan, HTTP redirect check	testssl.sh	SR-02	TC-16	Only TLS 1.2+; HTTP redirected
Authorization	Horizontal and vertical privilege-escalation tests	Postman, Burp Suite	SR-03	TC-12	Forbidden calls return 403
Session management	Token expiry and logout replay tests	Postman	SR-04	TC-22	Expired/logged-out token $\rightarrow$ 401
Input validation	SQL injection and XSS tests	OWASP ZAP, manual	SR-05	TC-11	No injection; output encoded
Data exposure	Inspect API payloads for answer keys	Browser dev tools, proxy	SR-07	TC-13	No answer key before submission
Exam integrity	Late-submission and timer-tampering tests	Postman	SR-09	TC-08	Late answers rejected
Audit trail	Verify entries and immutability	Admin UI, API	SR-08, FR-21	TC-19	All events logged; no edit/delete
Vulnerability scan	Baseline + active scan; dependency scan	ZAP, Dependency-Check	SR-10	TC-23	No open High/Critical findings

5.2 Functional Testing

Black-box techniques are used: equivalence partitioning and boundary-value analysis (for example duration 4, 5, 240 and 241 minutes; password lengths 7 and 8), decision tables (exam eligibility) and state-transition testing (attempt states: NOT_STARTED $\rightarrow$ IN_PROGRESS $\rightarrow$ SUBMITTED).

5.3 Performance and Load Testing

Apache JMeter simulates 500 and 1,000 concurrent students performing login, exam start, answer saving and submission. Response time percentiles, throughput, error rate and CPU/memory usage are recorded and compared with NFR-01, NFR-02 and NFR-04.

5.4 Usability, Compatibility and Accessibility Testing

A moderated usability session with five first-time users measures task completion time (NFR-05). The main flows are run on Chrome, Firefox, Edge and Safari and at 360 px viewport width (NFR-06). axe-core and keyboard-only navigation check WCAG 2.1 AA conformance (NFR-10).

5.5 Inspection-based Verification

- NFR-03 Availability: review of monitoring dashboard/uptime records during the mock exam period.
- NFR-08 Maintainability: unit-test coverage report from the build pipeline ($\geq 70\%$).
- NFR-09 Recoverability: a backup and restore drill timed against $RPO \leq 24$ h and $RTO \leq 4$ h.

5.6 Regression Testing

After every defect fix, the affected test cases and all High-priority test cases are re-run. The automated API suite runs on each commit.

6. Pass/Fail and Suspension Criteria

- A test case **passes** when the actual result matches the expected result with no unexpected error; otherwise it **fails** and a defect is logged.
- Testing is **suspended** if the build cannot be deployed, the login function is unusable, or more than 30 % of executed test cases fail in one cycle.
- Testing **resumes** after a fixed build passes the smoke test (TC-01, TC-02, TC-09).

7. Test Deliverables

- This Test Plan and the test case specification (Section 12).
- Test execution log and defect reports (tracked in the project's bug-tracking system).
- Load-test report, security scan reports (ZAP, testssl.sh, Dependency-Check).
- Test summary report with the traceability matrix showing execution status.

8. Test Environment

Component	Specification
Test server	Linux VM, 4 vCPU, 8 GB RAM; Docker; Java 17, Spring Boot 3, PostgreSQL 15, Redis 7
Client devices	Windows/macOS laptops, Android phone (360 px viewport)
Browsers	Latest two versions of Chrome, Firefox, Edge, Safari
Tools	Postman/Newman, Selenium, JMeter, OWASP ZAP, Burp Suite Community, testssl.sh, axe-core, JUnit
Data	Seeded users (3 roles), 50 questions, 3 exams, 10 student accounts

9. Roles, Responsibilities and Schedule

Role	Responsibility
Test lead	Owens the plan, assigns test cases, reports status.
Testers / developers	Write and execute test cases, log defects, fix and retest.
Instructor / reviewers	Review plan and approve exit criteria.

Activity	Planned period
Test planning and case design	24–28 Sep 2026 (this deliverable)
Unit and integration testing	During implementation (Part-2)
System, security and performance testing	After feature freeze

Activity	Planned period
Acceptance test and summary report	Before final submission

10. Risks and Contingencies

Risk	Impact	Mitigation
Load-test environment smaller than production	Performance results not representative	Scale results using per-instance throughput; repeat with 2 instances (TC-10)
Late changes to requirements	Test cases become outdated	Maintain traceability matrix; update affected cases immediately
Limited testing time	Low coverage of Low-priority items	Risk-based ordering: High-priority and security cases first
Time-dependent cases (timer, expiry) are slow	Delays in execution	Use configurable short durations and controllable clock in the test environment

11. Test Data and Tools

Test data is synthetic; no real student data is used. Passwords in the samples are for testing only. Tools are listed in Section 8. Each test case lists its own input data in the Test Data column of Section 12.

12. Test Cases

The following 23 test cases cover functional, non-functional and security requirements. Type: Functional / Non-functional / Security. All test cases are traceable to SRS requirements (Section 13).

ID	Title / Requirement / Type	Preconditions and steps	Test data	Expected result
TC-01	Register a new student with valid data Req: FR-01 Type: Functional Priority: High	Pre: e-mail not yet registered. 1. Open Register page. 2. Enter name, e-mail and a compliant password. 3. Click Register.	Name: Asha K; Email: asha@test.com; Password: Exam@2026	Account created; confirmation shown; user can log in. Duplicate e-mail or password “abc” is rejected with a field error.
TC-02	Login with valid credentials and role redirect Req: FR-02 Type: Functional Priority: High	Pre: student, teacher and admin accounts exist. 1. Open Login page. 2. Enter valid credentials. 3. Click Login and measure time.	student1 / teacher1 / admin1 with valid passwords	Each user lands on the correct dashboard within 3 s; invalid password shows “Invalid email or password”.
TC-03	Account lockout and login rate limit Req: FR-03, SR-06 Type: Security Priority: High	Pre: active student account. 1. Submit a wrong password 5 times within 1 minute. 2. Submit the correct password. 3. Send a 6th login request from same IP in the same minute.	student1 with wrong password “x1”	Account is locked for 15 min (correct password refused with lock message); the 6th request returns HTTP 429.
TC-04	Password reset link validity Req: FR-04 Type: Functional Priority: Medium	1. Click “Forgot password” and enter registered e-mail. 2. Open the link received; set a new password. 3. Re-use the same link. 4. Request a new link and use it after 31 minutes.	Registered e-mail; new password Reset@2026	First use succeeds; re-used link and link older than 30 min are rejected as invalid/expired.
TC-05	Teacher creates an exam Req: FR-06, FR-08 Type: Functional Priority: High	Pre: teacher logged in; 10 questions in the bank. 1. Open Exam Editor. 2. Enter title, duration, window, marks. 3. Add 10 questions; set negative marking 25 %. 4. Save.	Duration 30 min; total 10 marks; pass 4	Exam saved in DRAFT state with 10 questions; duration outside 5–240 min is rejected.

ID	Title / Requirement / Type	Preconditions and steps	Test data	Expected result
TC-06	Manage question bank (add / edit / delete) Req: FR-07 Type: Functional Priority: High	Pre: teacher logged in. 1. Add an MCQ with 4 options and 1 correct. 2. Add a True/False question. 3. Edit marks of the MCQ. 4. Delete the True/False question.	MCQ: “2+2 = ?” options 3,4,5,6; correct 4; marks 1	Questions are created, updated and deleted; an MCQ with no correct option cannot be saved.
TC-07	Exam access only inside the schedule window and attempt limit Req: FR-11 Type: Functional Priority: High	Pre: published exam window 10:00–11:00; attempt limit 1. 1. Try to start at 09:55. 2. Start at 10:05 and submit. 3. Try to start again. 4. Try again after 11:00 with another student.	Two student accounts; system time controlled	Start refused before 10:00 and after 11:00 with a reason; second attempt refused after the first is submitted.
TC-08	Timer auto-submit and server-side timing Req: FR-12, SR-09 Type: Functional Priority: High	Pre: exam of 5 minutes; student attempting. 1. Answer 3 questions; do nothing further. 2. Wait for the timer to reach 00:00. 3. Send an answer via API 15 s after end time.	Duration 5 min	Attempt is auto-submitted at 00:00 with saved answers; the late API call is rejected (HTTP 403/409).
TC-09	Automatic evaluation, negative marking and result view Req: FR-08, FR-15, FR-16, FR-17 Type: Functional Priority: High	Pre: exam with 10×1 -mark questions, negative marking 25 %. 1. Answer 6 correct, 2 wrong, 2 unanswered. 2. Click Submit and confirm. 3. Open Result page; time the display.	Answer set A (6C/2W/2U)	Score = $6 - (2 \times 0.25) = 5.5$ / 10; percentage 55 %; status per passing marks; result shown within 5 s; attempt is read-only.
TC-10	Load and performance test Req: NFR-01, NFR-02, NFR-04 Type: Non-functional Priority: High	Pre: staging environment; JMeter script for login → start exam → answer → submit. 1. Ramp to 500 virtual users over 5 min; hold 15 min. 2. Repeat with 1,000 users and 2 app instances. 3. Measure 95th-percentile latency and result-computation time (100-question exam).	500 and 1,000 virtual users; 100-question exam	95 % of requests ≤ 2 s at 500 users; result computed ≤ 5 s; error rate < 1 %; 1,000 users handled with 2 instances.
TC-11	SQL injection and XSS on inputs Req: SR-05 Type: Security Priority: High	1. On the login page enter the e-mail admin@x.com OR '1'='1 with any password. 2. In question text enter <code><script>alert(1)</script></code> and save. 3. View the question as a student. 4. Run OWASP ZAP active scan on these forms.	Payloads: ' OR '1'='1 ; <script>alert(1)</script>	Login fails; script is stored as inert text and not executed; no injection alert in the ZAP report.
TC-12	Role-based access control enforcement Req: SR-03 Type: Security Priority: High	Pre: student token available. 1. Call POST /api/v1/exams with student token. 2. Call GET /api/v1/admin/users with student token. 3. Call GET /api/v1/attempts/{id} of another student.	Student JWT; other student's attempt id	All three calls return HTTP 403 and an audit entry is written; no data is returned.
TC-13	Answer key not exposed before submission Req: SR-07 Type: Security Priority: High	1. Start an exam attempt. 2. Capture the response of the start/question API in the browser dev tools or a proxy. 3. Search the payload for correct-answer fields.	Any published exam	The payload contains no correct-answer or explanation fields until the attempt is submitted.
TC-14	Usability, browser compatibility and accessibility Req: NFR-05, NFR-06, NFR-10 Type: Non-functional Priority: Medium	1. Give 5 first-time students the task “log in and start the demo exam”; time each. 2. Run the main flow on Chrome, Firefox, Edge, Safari and at 360 px width. 3. Run axe accessibility scan and keyboard-only navigation.	5 volunteers; 4 browsers; axe-core	≥ 90 % complete the task within 3 min; no layout break on all browsers/360 px; no critical axe violations; contrast $\geq 4.5:1$.
TC-15	Autosave and recovery after connection loss Req: FR-13, NFR-07 Type: Non-functional Priority: High	Pre: attempt in progress. 1. Answer 5 questions and wait 35 s. 2. Disable the network for 3 minutes. 3. Re-enable network and refresh the page.	Network throttling tool	Autosave occurs at least every 30 s; on return the attempt resumes with the 5 saved answers and the correct remaining time.

ID	Title / Requirement / Type	Preconditions and steps	Test data	Expected result
TC-16	Password hashing and transport encryption Req: SR-01, SR-02 Type: Security Priority: High	1. Register a user and inspect the users table. 2. Open http:// URL of the site. 3. Run testssl.sh against the host.	Test user; staging host	Password column contains a bcrypt hash (cost ≥ 12), never plaintext; HTTP redirects to HTTPS; only TLS 1.2/1.3 are enabled.
TC-17	Administrator manages users Req: FR-05 Type: Functional Priority: High	Pre: admin logged in. 1. Create a teacher account. 2. Edit its role to Student. 3. Deactivate it and try to log in as that user. 4. Delete the account.	New user teacher2@test.com	Each action succeeds and is reflected in the list; the deactivated user cannot log in.
TC-18	Publish exam and student visibility Req: FR-09, FR-10 Type: Functional Priority: High	Pre: draft exam assigned to Group A. 1. Student of Group A opens dashboard (exam hidden). 2. Teacher publishes the exam. 3. Student refreshes; student of Group B checks.	Groups A and B	Exam appears under Upcoming/Available only for Group A after publishing; not visible before publishing or to Group B.
TC-19	Audit log completeness and immutability Req: FR-21, SR-08 Type: Security Priority: Medium	1. Perform login, logout, exam start, submit and a user edit. 2. As admin open the audit log. 3. As teacher try to open the audit log. 4. Try to edit/delete a log row via API.	Actions from earlier steps	Every action appears with user, action, timestamp and IP; non-admin access returns 403; modification of entries is not possible.
TC-20	Result reports and CSV export Req: FR-18, FR-19 Type: Functional Priority: Medium	Pre: 10 submitted attempts. 1. Teacher opens Reports for the exam. 2. Verify average/highest/lowest/pass %. 3. Click Export CSV. 4. Admin opens summary report.	10 attempts with known scores	Statistics equal manual calculation; CSV has one row per student; admin totals of users/exams/attempts are correct.
TC-21	Question randomization and mark-for-review Req: FR-14, FR-20 Type: Functional Priority: Medium	Pre: exam with randomization on. 1. Two students start the same exam. 2. Compare question order. 3. One student marks Q3 for review and jumps to Q8 and back.	Two student accounts	Question/option order differs between students; review flag and previously chosen answer are retained on navigation.
TC-22	Token expiry and logout invalidation Req: SR-04 Type: Security Priority: High	1. Log in and copy the JWT. 2. Wait 31 minutes idle, call an API with it. 3. Log in again, log out, replay the old token.	JWT of student1	Expired token → HTTP 401; token used after logout → HTTP 401.
TC-23	OWASP Top 10 scan of release build Req: SR-10 Type: Security Priority: Medium	1. Deploy the release candidate to staging. 2. Run OWASP ZAP baseline + active scan. 3. Run dependency-vulnerability scan. 4. Review results.	ZAP, OWASP Dependency-Check	No open High or Critical findings; Medium findings have a documented mitigation.

13. Traceability Matrix

13.1 Functional Requirements

Requirement	Summary	Test case(s)
FR-01	Allow a new student to register by providing full name, a unique email address and a password t...	TC-01
FR-02	Authenticate a user with email and password and shall redirect the user to the dashboard for hi...	TC-02
FR-03	Temporarily lock a user account for 15 minutes after 5 consecutive failed login attempts...	TC-03
FR-04	Allow a user to reset a forgotten password through a single-use link sent to the registered ema...	TC-04
FR-05	Allow an Administrator to create, edit, deactivate and delete user accounts and to assign one r...	TC-17

Requirement	Summary	Test case(s)
FR-06	Allow a Teacher to create an exam by specifying title, subject, duration (5–240 minutes), start...	TC-05
FR-07	Allow a Teacher to add, edit and delete questions in a question bank...	TC-06
FR-08	Allow a Teacher to select questions from the question bank for an exam and to configure negativ...	TC-05, TC-09
FR-09	Allow a Teacher to publish or unpublish an exam and to assign it to one or more student groups...	TC-18
FR-10	Display to a Student the list of assigned exams grouped as Upcoming, Available and Completed...	TC-18
FR-11	Allow a Student to start an exam only between its start and end date-time and only if the numbe...	TC-07
FR-12	Display a countdown timer during an attempt and shall automatically submit the attempt when the...	TC-08
FR-13	Automatically save the answers of an in-progress attempt at least once every 30 seconds...	TC-15
FR-14	Allow a Student to move to any question, change an answer, and mark a question for review befor...	TC-21
FR-15	Allow a Student to submit an attempt manually after a confirmation prompt...	TC-09
FR-16	Automatically evaluate all objective answers on submission and shall compute the total score ap...	TC-09
FR-17	Show the Student his/her score, percentage and Pass/Fail status within 5 seconds of submission,...	TC-09
FR-18	Allow a Teacher to view result reports of his/her exams (average, highest, lowest, pass percent...	TC-20
FR-19	Allow an Administrator to view a summary report of the total number of users, exams and attempt...	TC-20
FR-20	Randomize the order of questions and options per Student when the Teacher enables randomization...	TC-21
FR-21	Record an audit log entry (user, action, timestamp, IP address) for every login, logout, exam s...	TC-19

13.2 Non-Functional Requirements

Requirement	Category	Test case(s)
NFR-01	Performance	TC-10
NFR-02	Performance	TC-10
NFR-03	Availability	Inspection (5.5)
NFR-04	Scalability	TC-10
NFR-05	Usability	TC-14
NFR-06	Compatibility	TC-14
NFR-07	Reliability	TC-15
NFR-08	Maintainability	Inspection (5.5)
NFR-09	Recoverability	Inspection (5.5)
NFR-10	Accessibility	TC-14

13.3 Security Requirements

Requirement	Objective(s)	Test case(s)
SR-01	SO-1	TC-16

Requirement	Objective(s)	Test case(s)
SR-02	SO-1	TC-16
SR-03	SO-1, SO-2	TC-12
SR-04	SO-5	TC-22
SR-05	SO-2	TC-11
SR-06	SO-3	TC-03
SR-07	SO-1, SO-5	TC-13
SR-08	SO-4	TC-19
SR-09	SO-2, SO-5	TC-08
SR-10	SO-1, SO-2, SO-3	TC-23

Coverage summary: 21 of 21 functional requirements, 7 of 10 non-functional requirements by test and 3 by inspection, and 10 of 10 security requirements are covered. Total test cases: 23.

14. Approvals

Role	Name	Signature	Date
Test lead	[Name]		
Project guide / Instructor	[Name]		